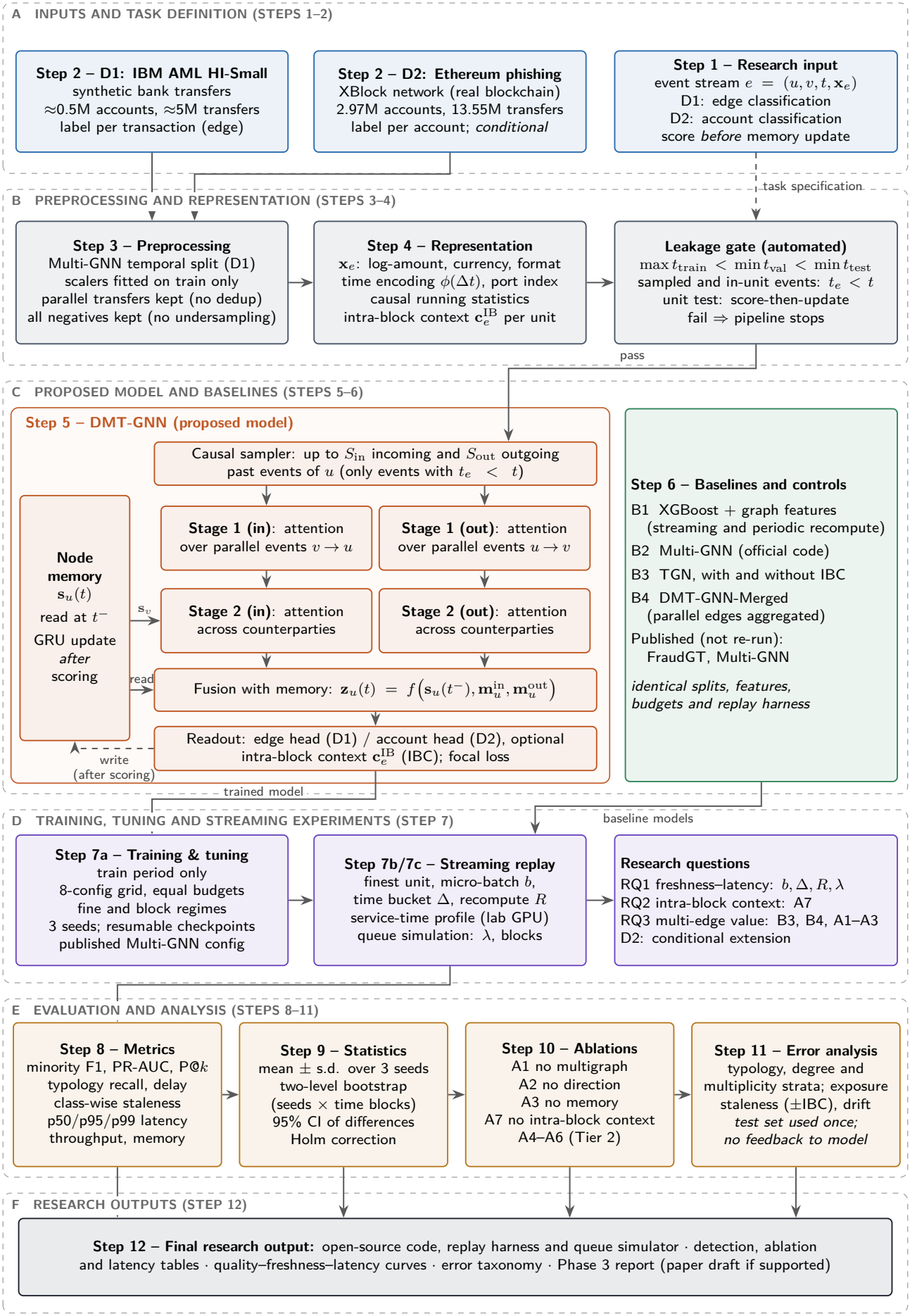

A INPUTS AND TASK DEFINITION (STEPS 1–2)
Step 2 – D1: IBM AML HI-Small
synthetic bank transfers
≈0.5M accounts, ≈5M transfers
label per transaction (edge)
Step 2 – D2: Ethereum phishing
XBlock network (real blockchain)
2.97M accounts, 13.55M transfers
label per account; conditional
Step 1 – Research input
event stream $e = (u, v, t, \mathbf{x}_e)$
D1: edge classification
D2: account classification
score before memory update
task specification
B PREPROCESSING AND REPRESENTATION (STEPS 3–4)
Step 3 – Preprocessing
Multi-GNN temporal split (D1)
scalers fitted on train only
parallel transfers kept (no dedup)
all negatives kept (no undersampling)
Step 4 – Representation
$\mathbf{x}_e$: log-amount, currency, format
time encoding $\phi(\Delta t)$, port index
causal running statistics
intra-block context $\mathbf{c}_e^{\mathrm{IB}}$ per unit
Leakage gate (automated)
$\max t_{\mathrm{train}} < \min t_{\mathrm{val}} < \min t_{\mathrm{test}}$
sampled and in-unit events: $t_e < t$
unit test: score-then-update
fail ⇒ pipeline stops
pass
C PROPOSED MODEL AND BASELINES (STEPS 5–6)
Step 5 – DMT-GNN (proposed model)
Causal sampler: up to S_{in} incoming and S_{out} outgoing past events of u (only events with $t_e < t$)
Node memory
$\mathbf{s}_u(t)$
read at t^-
GRU update after scoring
$\mathbf{s}_v$
read
write (after scoring)
Stage 1 (in): attention over parallel events $v \to u$
Stage 1 (out): attention over parallel events $u \to v$
Stage 2 (in): attention across counterparties
Stage 2 (out): attention across counterparties
Fusion with memory: $\mathbf{z}_u(t) = f\left(\mathbf{s}_u(t^-), \mathbf{m}_u^{\mathrm{in}}, \mathbf{m}_u^{\mathrm{out}}\right)$
Readout: edge head (D1) / account head (D2), optional intra-block context $\mathbf{c}_e^{\mathrm{IB}}$ (IBC); focal loss
trained model
Step 6 – Baselines and controls
B1 XGBoost + graph features (streaming and periodic recompute)
B2 Multi-GNN (official code)
B3 TGN, with and without IBC
B4 DMT-GNN-Merged (parallel edges aggregated)
Published (not re-run): FraudGT, Multi-GNN
identical splits, features, budgets and replay harness
baseline models
D TRAINING, TUNING AND STREAMING EXPERIMENTS (STEP 7)
Step 7a – Training & tuning
train period only
8-config grid, equal budgets
fine and block regimes
3 seeds; resumable checkpoints
published Multi-GNN config
Step 7b/7c – Streaming replay
finest unit, micro-batch b,
time bucket Δ, recompute R
service-time profile (lab GPU)
queue simulation: λ, blocks
Research questions
RQ1 freshness–latency: b, Δ, R, λ
RQ2 intra-block context: A7
RQ3 multi-edge value: B3, B4, A1–A3
D2: conditional extension
E EVALUATION AND ANALYSIS (STEPS 8–11)
Step 8 – Metrics
minority F1, PR-AUC, P@k
typology recall, delay
class-wise staleness
p50/p95/p99 latency
throughput, memory
Step 9 – Statistics
mean ± s.d. over 3 seeds
two-level bootstrap
(seeds × time blocks)
95% CI of differences
Holm correction
Step 10 – Ablations
A1 no multigraph
A2 no direction
A3 no memory
A7 no intra-block context
A4–A6 (Tier 2)
Step 11 – Error analysis
typology, degree and multiplicity strata; exposure staleness (±IBC), drift
test set used once; no feedback to model
F RESEARCH OUTPUTS (STEP 12)
Step 12 – Final research output: open-source code, replay harness and queue simulator · detection, ablation and latency tables · quality–freshness–latency curves · error taxonomy · Phase 3 report (paper draft if supported)